

RUBRICWORLD: World Model-Guided Rubric Optimization for LLM Graders

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Abstract

In rubric optimization, each edit acts on an LLM grader’s current scoring rubric, potentially correcting grading errors while degrading previously correct decisions. The same edit can also produce different effects as earlier revisions alter the current rubric state. Existing prompt and rubric optimizers commonly select edits using aggregate validation performance, without explicitly learning these state-dependent effects. We introduce RUBRICWORLD, a world model-guided framework for state-conditioned rubric optimization. Learning from paired grading decisions before and after each edit, RUBRICWORLD predicts correction and degradation rates at ordinal score boundaries to guide candidate selection. Independent empirical certification then uses held-out measurements and finite-sample confidence bounds to determine whether an edit is committed. Across EIR (National), ASAP2.0, and ASAP-SAS with three frozen LLM graders, RUBRICWORLD achieves the best results in 20 of 27 reported metric comparisons. On ASAP2.0, RUBRICWORLD increases gpt-5-mini accuracy from 18.20% to 46.70%. End-to-end ablations further show that removing rubric-state information reduces average accuracy from 40.18% to 35.42%, while removing the learned edit-effect model reduces it to 27.31%. These results support learning state-dependent edit effects to guide rubric optimization.

1 Introduction

Automatic scoring has a long history, from feature-based essay graders to learned scoring systems (Page, 1966; Attali and Burstein, 2006; Taghipour and Ng, 2016). Large language models (LLMs) extend this work to open-ended answers assessed using natural-language instructions (Mansour et al., 2024; Henkel et al., 2024). LLM evaluators are also used more broadly for assessing generated responses (Zheng et al., 2023; Vu et al., 2024).

These instructions can include rubrics, scored examples, and explicit criteria for awarding each score (Hashemi et al., 2024; Zhou et al., 2025). Since a scoring rubric is written in natural language, it can be revised without retraining the grader (Chu et al., 2025, 2026). The changes remain inspectable: teachers can review an amended rule, reject it, or revise it directly (Henkel et al., 2025). This makes rubric editing practical, but repeated identification and correction of grading errors is costly (Burrows et al., 2015; Mohler et al., 2011; Sakaguchi et al., 2015).

Rubric and prompt optimizers such as APO, GradeOpt, CARO, and GEPA generate revisions from errors, feedback, or evaluation traces (Pryzant et al., 2023; Chu et al., 2025, 2026; Agrawal et al., 2026). Their search often emphasizes aggregate validation performance rather than learning the paired effect of an edit in the current rubric state (Yang et al., 2024; Agrawal et al., 2026). Aggregate scores can conceal newly incorrect decisions even as overall agreement improves, especially under prompt sensitivity and judge bias (Sclar et al., 2024; Chen et al., 2024; Dorner et al., 2025).

One rubric edit can correct some decisions while degrading others. Suppose an edit makes the requirement for Score 4 more strict. It may correctly lower a group of over-scored answers from 4 to 3, while also lowering answers that genuinely deserve 4. If the first group is larger, overall accuracy can still increase. Aggregate performance therefore summarizes the net effect of the edit without showing which previously correct judgments were degraded. Ambiguous scoring criteria can weaken both agreement and score meaning (Kane, 2013; Henkel et al., 2025; Messick, 1995).

The ordered score scale provides a way to locate these changes through ordinal thresholds (McCullagh, 1980; Cao et al., 2020). A prediction of score 4, for example, contains decisions about whether an answer lies above the 1|2, 2|3, and 3|4 ordinal

score boundaries. An edit intended to correct errors near $3/4$ may also change decisions at $2/3$ or $4/5$. We therefore measure an edit by comparing grading decisions on the same answers before and after it is applied. The resulting *edit effect* records both corrections to previously incorrect decisions and degradations of previously correct decisions, including where these changes occur on the ordered score scale.

Repeated editing adds a second difficulty: later edits are applied to rubrics altered by earlier rounds, as in other iterative instruction refinement procedures (Yang et al., 2024; Madaan et al., 2023). The same edit may work differently after a rule has been added, removed, or strengthened; prompt sensitivity provides a reason to expect such variation (Sclar et al., 2024; Pu et al., 2026). Thus, an edit’s effect depends on both the edit and the current rubric state. Re-evaluating every candidate is costly, whereas paired effects from earlier rounds may offer reusable evidence for selecting the next edit (Yang et al., 2024; Opsahl-Ong et al., 2024).

RUBRICWORLD addresses these two properties jointly. Figure 1 outlines the full generate–predict–measure–certify loop. At each round, a candidate generator first localizes an observed grading failure and produces a small set of scoped text edits with a specified correction direction. A state-conditioned edit-effect model, motivated by action-conditioned prediction (Hansen et al., 2024; Puterman, 1994), estimates how each candidate would change grading decisions under the current state. The model represents an edit’s state-independent effect together with its interaction with the current state, allowing the same candidate to have different predicted effects at different stages of optimization. These predictions are used to select one candidate for paired evaluation. The selected edit is applied to the frozen grader on the same answers before and after editing, and its measured effect is retained to improve later predictions.

The final commit decision is made independently of the edit-effect model. The selected edit is evaluated on held-out answers not used for construction, training, or selection; separating adaptive search from confirmation limits feedback reuse (Dwork et al., 2015; Alagharu et al., 2026). We use one-sided finite-sample confidence bounds on target correction and degradation to decide whether the edit meets predeclared limits (Clopper and Pearson, 1934). The rule retains the current rubric whenever the held-out evidence cannot support the required

certification bounds. Thus, learned edit effects determine which candidate is evaluated, while held-out evidence determines whether that candidate is committed to the rubric.

The value of modeling edits is reflected in our end-to-end ablations: removing rubric-state information lowers average grading accuracy, and removing the learned edit-effect model lowers it further. Our long-horizon analysis also tracks regressions that remain after subsequent edits. These results motivate predicting edit effects during optimization rather than judging only the final rubric.

Our experiments address five linked questions: how the same edit behaves across rubric states; whether its effects can be predicted to select better edits; whether held-out evidence supports committing an update; how the complete method compares in grading quality and token cost; and which regressions remain after repeated editing.

Our contributions are:

- **State-dependent rubric optimization.** We examine how rubric edits produce different correction and degradation effects across optimization states.
- **World model-guided optimization.** We propose RUBRICWORLD to learn edit effects, guide candidate selection, and independently certify updates.
- **Extensive empirical validation.** We evaluate diverse benchmarks and LLM graders, including ablations, optimization costs, and long-horizon behavior.

2 Related Work

Rubric-Guided LLM Evaluation. G-Eval uses task-specific evaluation steps to guide LLM scoring, while Prometheus learns fine-grained evaluation from supplied rubrics (Liu et al., 2023; Kim et al., 2024a). LLM-Rubric further combines and calibrates criterion-level judgments for more structured evaluation (Hashemi et al., 2024). These methods primarily study how a given scoring rubric is applied or calibrated, rather than how it should be revised repeatedly from observed grading errors (Kim et al., 2024a; Hashemi et al., 2024).

Feedback-Driven Prompt and Rubric Optimization. APO and GEPA improve instructions using textual feedback or evaluation traces, while GradeOpt derives grading guidelines from misgraded answers (Pryzant et al., 2023; Agrawal et al.,

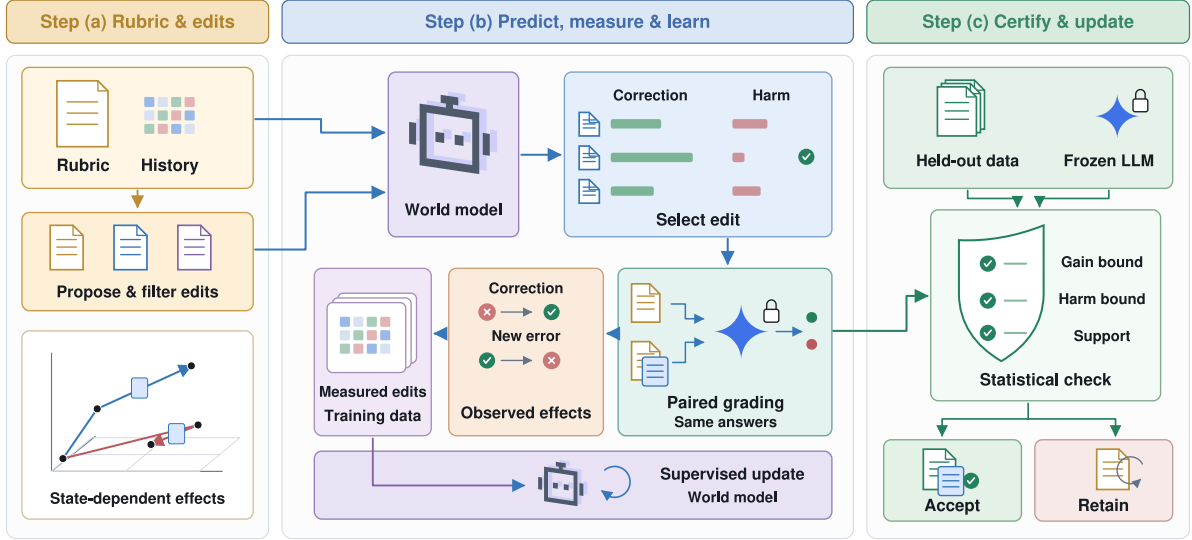


Figure 1: **Overview of RUBRICWORLD.** (a) Boundary-targeted edits are proposed from the current rubric and errors. (b) A state-conditioned world model predicts repair and harm, guides selection, and learns from paired transitions, including rejected edits. (c) Held-out confidence bounds govern rubric updates. The LLM grader is frozen.

2026; Chu et al., 2025). PromptAgent formulates prompt refinement as sequential planning, whereas MIPRO learns a surrogate objective to guide candidate search (Wang et al., 2024; Opsahl-Ong et al., 2024). Recent optimizers also introduce preservation or degradation constraints to reduce losses on previously correct behavior (Xu et al., 2026; Satheesha et al., 2026). These approaches improve candidate generation, search, or update selection, but do not explicitly learn the paired edit effect conditioned on the current rubric state; this matters because earlier revisions change the state under which later edits are applied (Wang et al., 2024; Opsahl-Ong et al., 2024).

World Model Learning. Recent world-model methods learn action-conditioned dynamics and use predicted outcomes to guide subsequent planning or control (Hansen et al., 2024; Hafner et al., 2025). RUBRICWORLD adopts this predictive view at the level of a single rubric edit: rather than predicting only a scalar prompt score, it models paired corrections and degradations across score boundaries under the current rubric state. This is one-step edit-effect prediction, not full next-state simulation. Predicted effects guide selection; separate held-out evidence decides whether the edit is committed.

3 Methodology

Figure 1 illustrates the three-stage optimization pipeline and its paired-effect learning feedback.

Given a current scoring rubric, RUBRICWORLD predicts how candidate text edits would change the behavior of a frozen LLM grader. These predicted edit effects guide candidate selection; the selected edit is then measured from paired before–after grading decisions and independently certified before it can be committed. Observed edit effects are retained to improve prediction in later rounds. The update rule and its effect measurements are detailed below.

3.1 Problem Formulation and Data Setting

Let f denote a frozen LLM grader and let R_t denote the scoring rubric at optimization round t . For a graded answer x_i with human reference score y_i ,

$$\hat{y}_i^{(t)} = F_f(x_i; R_t).$$

A candidate edit p changes the rubric to $R_t \oplus p$; the parameters of f remain fixed.

For K consecutive score labels $y_{\min}, \dots, y_{\max}$, let $\mathcal{B} = \{y_{\min}, \dots, y_{\max} - 1\}$ index the $K - 1$ ordinal thresholds (McCullagh, 1980; Cao et al., 2020); b separates scores b and $b + 1$ (e.g., $0 \mid 1$ for an EIR item scored 0–3). At round t , candidate generation selects a target boundary $b_t^* \in \mathcal{B}$ and produces

$$\mathcal{P}_t = \{p_t^1, \dots, p_t^M\}.$$

For an edit p , let $C_{t,b}(p)$ denote its correction rate at boundary b , i.e., the fraction of currently incorrect decisions that become correct. Let $D_{t,b}(p)$

denote its degradation rate, i.e., the fraction of currently correct decisions that become incorrect. For each boundary, ϵ_b specifies the maximum tolerated degradation rate. The ideal round-level objective is

$$\begin{aligned} p_t^* &\in \arg \max_{p \in \mathcal{P}_t} C_{t,b_t^*}(p) \\ \text{s.t. } D_{t,b}(p) &\leq \epsilon_b, \quad \forall b \in \mathcal{B}. \end{aligned} \quad (1)$$

This boundary-level objective controls corrections and regressions, but is a proxy rather than a direct optimizer of test QWK or ACC (Cohen, 1968). Accepted edits affect the rubric used in later rounds. We represent the current state as $s_t = (R_t, \mathcal{H}_t, \mathbf{v}_t)$, where $\mathcal{H}_t$ records earlier edit effects and $\mathbf{v}_t$ summarizes training-answer features, observed scores, and correctness, with probability and margin features where available. Only R_t is sent to the grader; the predictor does not require API-model logits.

Data setting. Four disjoint sets serve model learning ($\mathcal{D}_{\text{dyn,train}}$), model selection ($\mathcal{D}_{\text{dyn,val}}$), independent certification ($\mathcal{D}_{\text{cert}}$), and final testing ($\mathcal{D}_{\text{test}}$). Candidate construction and paired-effect training use only the training set; grouped evaluation by edit identity, edit text, and rubric state is distinct from these data splits (Appendix H.1).

3.2 Paired Edit Effects

We measure the effect of an edit by grading the same answers before and after it is applied. Because identical score changes can affect different parts of an ordered scale, we represent each score by its $K - 1$ threshold decisions:

$$z_{i,b} = \mathbb{I}(y_i > b), \quad \hat{z}_{i,b}^{(t)} = \mathbb{I}(\hat{y}_i^{(t)} > b).$$

Let $\hat{z}_{i,b}^p$ denote the corresponding decision under $R_t \oplus p$.

For each boundary, let

$$\mathcal{F}_{t,b} = \{i : z_{i,b} \neq \hat{z}_{i,b}^{(t)}\}$$

contain decisions that are incorrect before the edit, and let

$$\mathcal{A}_{t,b} = \{i : z_{i,b} = \hat{z}_{i,b}^{(t)}\}$$

contain decisions that are correct.

For nonempty $\mathcal{F}_{t,b}$, the correction rate is

$$C_{t,b}(p) = \frac{1}{|\mathcal{F}_{t,b}|} \sum_{i \in \mathcal{F}_{t,b}} \mathbb{I}(z_{i,b} = \hat{z}_{i,b}^p).$$

For nonempty $\mathcal{A}_{t,b}$, the degradation rate is

$$D_{t,b}(p) = \frac{1}{|\mathcal{A}_{t,b}|} \sum_{i \in \mathcal{A}_{t,b}} \mathbb{I}(z_{i,b} \neq \hat{z}_{i,b}^p).$$

We collect these measurements into the edit-effect vector

$$\mathbf{e}_t(p) = [\mathbf{C}_t(p); \mathbf{D}_t(p)]. \quad (2)$$

where $\mathbf{C}_t(p) = [C_{t,b}(p)]_{b \in \mathcal{B}}$ and $\mathbf{D}_t(p) = [D_{t,b}(p)]_{b \in \mathcal{B}}$.

$C_{t,b_t^*}(p)$ measures correction at the intended score boundary, while the remaining coordinates record effects elsewhere on the ordered score scale. Incomplete per-answer boundary records are dropped, never padded; undefined rate coordinates are masked (Appendix I.3).

3.3 State-Conditioned Edit-Effect Model

Let $a_p = \phi(p)$ encode candidate edit p and $h_t = \psi(s_t)$ encode the current state. The reported method uses hybrid edit features (text and Doc-to-LoRA (D2L) sketches (Charakorn et al., 2026)); the state encoder uses R_t , $\mathcal{H}_t$, and $\mathbf{v}_t$ (Appendix J.2).

We model the expected edit effect as

$$\boldsymbol{\eta}_\theta(s_t, p) = W_a a_p + U [(A a_p) \odot (B h_t)] + c, \quad (3)$$

followed by

$$\boldsymbol{\mu}_\theta(s_t, p) = \sigma(\boldsymbol{\eta}_\theta(s_t, p)), \quad (4)$$

where θ collects the trainable parameters W_a , A , B , U , and c , together with trainable encoder parameters when applicable.

The first term captures the effect associated with the edit itself. The second captures how that effect changes with the current rubric state: the edit and state are projected into a shared space, combined through element-wise multiplication, and mapped to the predicted correction and degradation coordinates. This factorized interaction, following the idea of low-rank feature interactions (Rendle, 2010), allows the same edit to have different predicted effects under different rubric states.

The model is trained on edit-effect observations $\mathcal{T}_t = \{(s_n, p_n, \mathbf{e}_n)\}_{n=1}^{N_t}$ using

$$\mathcal{L}(\theta) = \sum_{(s,p,\mathbf{e}) \in \mathcal{T}_t} \ell(\boldsymbol{\mu}_\theta(s, p), \mathbf{e}), \quad (5)$$

where the loss includes only effect coordinates with nonzero empirical support. A six-state pilot provides 7,200 complete answer–edit observations,

corresponding to 36,000 boundary-level records. Detailed encoders, grouped validation, and warm-start rules are in [Appendix J](#).

3.4 Model-Guided Candidate Selection

For candidate p_t^j , let $\mu_{t,j,b_t^*}^C$ and $\sigma_{t,j,b_t^*}^C$ denote the predicted mean and uncertainty of target correction. For boundary b , let $\mu_{t,j,b}^D$ and $\sigma_{t,j,b}^D$ denote the corresponding degradation quantities.

We construct an LCB-style conservative score for correction,

$$\tilde{C}_{t,j} = \mu_{t,j,b_t^*}^C - \beta \sigma_{t,j,b_t^*}^C,$$

and a UCB-style conservative score for degradation,

$$\tilde{D}_{t,j,b} = \mu_{t,j,b}^D + \beta \sigma_{t,j,b}^D,$$

where $\beta \geq 0$ penalizes predictive uncertainty (the standard deviation across five independently seeded models) ([Lakshminarayanan et al., 2017](#); [Guo et al., 2017](#)).

We select

$$p_t^* = \arg \max_{p_t^j \in \mathcal{P}_t} \tilde{C}_{t,j} \quad \text{s.t.} \quad \tilde{D}_{t,j,b} \leq \epsilon_b, \quad \forall b \in \mathcal{B}. \quad (6)$$

The LCB/UCB-style quantities above are predictive selection scores, not calibrated statistical confidence bounds. The selected candidate is then evaluated on $\mathcal{D}_{\text{dyn},\text{train}}$ using the paired procedure in [Section 3.2](#). Its measured edit effect is added to the edit-effect observations and used to update the model for later rounds.

3.5 Independent Empirical Certification

The final commit decision is made independently on held-out certification data. For the selected edit, let $L_{t,b_t^*}^C$ denote a one-sided lower confidence bound on target correction and $U_{t,b}^D$ a one-sided upper confidence bound on degradation.

Let γ denote the minimum certified correction rate, while ϵ_b remains the maximum tolerated degradation rate at boundary b . A candidate is certifiable only if $b_t^* \in \mathcal{B}_{\text{cert}}$, where $\mathcal{B}_{\text{cert}} \subseteq \mathcal{B}$ is the pre-specified certification scope. Predicted constraints screen all score boundaries, whereas finite-sample certification covers only this predeclared subset. The edit is committed only if

$$L_{t,b_t^*}^C \geq \gamma, \quad U_{t,b}^D \leq \epsilon_b \quad \forall b \in \mathcal{B}_{\text{cert}}. \quad (7)$$

These quantities are statistical confidence bounds computed from paired grader decisions using exact one-sided binomial intervals ([Clopper and Pearson, 1934](#)). Certification requires sufficient sample support for every prescribed bound.

A certified edit gives $R_{t+1} = R_t \oplus p_t^*$; otherwise $R_{t+1} = R_t$. Confidence allocation, minimum support, and decision cases are given in [Appendix L](#).

4 Experiments

4.1 Experimental Design

Five comparisons follow the editing process from individual effects to final grading. We examine final quality ([Table 1](#)), token cost ([Figure 3](#)), the effect of the starting rubric ([Figure 4](#)), the contribution of prediction, selection, and certification ([Figure 2](#) and [Table 2](#)), and the consequences of 100 successive rounds ([Figures 5](#) and [6](#)).

Benchmarks. EIR (National) contains five private short-answer items, with item-specific scores from 0 to 3 or 0 to 4 ([Chu et al., 2025](#)). ASAP2.0 contains student essays scored from 1 to 6 ([The Learning Agency Lab, 2024](#); [Crossley et al., 2025](#)). ASAP-SAS evaluates short answers; the reported Set 1 uses scores from 0 to 3 ([Barbara et al., 2012](#); [Sakaguchi et al., 2015](#)). The ordered score scales allow us to examine both exact scores and decisions at each adjacent score boundary. Counts and task-level splits are in [Appendix B](#).

Frozen LLM graders. We use Llama-3.1-8B-Instruct, an 8B-parameter instruction-tuned language model ([Grattafiori et al., 2024](#)); QwQ-32B-Preview, a 32B-parameter reasoning language model from Qwen ([Qwen Team, 2024](#)); and OpenAI’s API-based language model gpt-5-mini ([OpenAI, 2026](#)). Llama and QwQ use 4-bit activation-aware weight-quantized (AWQ) checkpoints ([Lin et al., 2024](#)) named in [Table 1](#). All three graders stay frozen; the scoring rubric is the only part updated at inference time.

4.2 Methods, Protocol, and Measures

Comparison methods. Zero-shot uses the original rubric, and few-shot adds labeled examples without changing it ([Brown et al., 2020](#)). APO edits instructions using language-based feedback on mistakes ([Pryzant et al., 2023](#)); GradeOpt refines grading guidelines from misgraded answers ([Chu et al., 2025](#)); and GEPA uses evaluated traces, reflection, and a Pareto-based prompt search ([Agrawal et al., 2026](#)). All methods begin with the same rubric and

Method	EIR (National)			ASAP2.0			ASAP-SAS		
	ACC	QWK	F1	ACC	QWK	F1	ACC	QWK	F1
Meta-Llama-3.1-8B-Instruct-AWQ-INT4									
Zero-shot (Brown et al., 2020)	55.42 $\pm$ 0.54	54.10 $\pm$ 0.46	54.85 $\pm$ 1.92	12.50 $\pm$ 2.15	5.75 $\pm$ 2.13	4.55 $\pm$ 0.70	1.99 $\pm$ 0.64	1.44 $\pm$ 0.87	1.20 $\pm$ 2.08
Few-shot (Brown et al., 2020)	57.12 $\pm$ 2.00	52.80 $\pm$ 0.56	55.15 $\pm$ 0.20	9.40 $\pm$ 0.54	4.78 $\pm$ 0.79	4.74 $\pm$ 1.45	19.92 $\pm$ 1.39	0.01 $\pm$ 2.25	7.75 $\pm$ 2.01
APO (Pryzant et al., 2023)	61.35 $\pm$ 1.17	60.15 $\pm$ 1.97	59.90 $\pm$ 1.17	15.45 $\pm$ 1.68	8.92 $\pm$ 1.27	7.98 $\pm$ 0.79	21.51 $\pm$ 0.75	8.90 $\pm$ 1.56	14.50 $\pm$ 0.61
GradeOpt (Chu et al., 2025)	54.20 $\pm$ 0.83	56.45 $\pm$ 0.18	53.10 $\pm$ 1.99	10.48 $\pm$ 0.68	7.37 $\pm$ 2.11	5.44 $\pm$ 1.53	33.07 $\pm$ 1.76	9.15 $\pm$ 1.53	21.09 $\pm$ 0.43
GEPA (Agrawal et al., 2026)	58.90 $\pm$ 2.09	57.20 $\pm$ 1.96	57.45 $\pm$ 0.31	13.10 $\pm$ 1.31	6.80 $\pm$ 1.62	5.90 $\pm$ 0.67	16.40 $\pm$ 0.96	4.50 $\pm$ 1.37	8.20 $\pm$ 1.62
RUBRICWORLD	64.88 $\pm$ 1.10	63.12 $\pm$ 0.45	62.45 $\pm$ 1.89	34.94 $\pm$ 0.52	32.94 $\pm$ 1.90	30.77 $\pm$ 0.30	29.50 $\pm$ 1.65	9.40 $\pm$ 1.22	20.15 $\pm$ 0.36
Qwen/QwQ-32B-Preview-AWQ-INT4									
Zero-shot (Brown et al., 2020)	68.15 $\pm$ 1.46	67.40 $\pm$ 2.20	67.90 $\pm$ 0.91	22.15 $\pm$ 0.72	18.40 $\pm$ 0.77	15.60 $\pm$ 1.29	14.20 $\pm$ 1.28	11.05 $\pm$ 0.43	10.80 $\pm$ 1.03
Few-shot (Brown et al., 2020)	66.40 $\pm$ 2.06	68.95 $\pm$ 2.10	65.20 $\pm$ 0.37	18.30 $\pm$ 2.14	12.60 $\pm$ 0.62	11.90 $\pm$ 1.64	25.40 $\pm$ 1.97	4.25 $\pm$ 0.24	15.60 $\pm$ 1.47
APO (Pryzant et al., 2023)	75.30 $\pm$ 1.10	71.85 $\pm$ 0.93	74.10 $\pm$ 1.26	31.40 $\pm$ 2.07	28.15 $\pm$ 1.63	25.90 $\pm$ 2.13	26.80 $\pm$ 1.65	14.30 $\pm$ 1.31	21.15 $\pm$ 2.02
GradeOpt (Chu et al., 2025)	70.12 $\pm$ 1.13	69.50 $\pm$ 2.05	69.95 $\pm$ 1.00	28.55 $\pm$ 2.11	22.10 $\pm$ 0.34	21.45 $\pm$ 1.87	33.45 $\pm$ 1.06	16.80 $\pm$ 0.35	28.40 $\pm$ 0.61
GEPA (Agrawal et al., 2026)	72.05 $\pm$ 1.54	69.80 $\pm$ 2.15	71.15 $\pm$ 2.20	26.80 $\pm$ 1.24	23.40 $\pm$ 0.20	20.25 $\pm$ 2.31	22.10 $\pm$ 1.92	12.50 $\pm$ 1.38	17.30 $\pm$ 0.68
RUBRICWORLD	73.85 $\pm$ 0.18	74.12 $\pm$ 1.96	72.60 $\pm$ 1.97	38.90 $\pm$ 1.35	37.25 $\pm$ 1.39	35.10 $\pm$ 0.76	34.12 $\pm$ 0.24	22.45 $\pm$ 0.62	31.15 $\pm$ 0.37
OpenAI gpt-5-mini									
Zero-shot (Brown et al., 2020)	73.42 $\pm$ 1.87	72.85 $\pm$ 0.50	73.11 $\pm$ 0.78	18.20 $\pm$ 1.18	13.55 $\pm$ 2.31	12.10 $\pm$ 0.30	7.80 $\pm$ 1.97	5.15 $\pm$ 0.97	5.45 $\pm$ 1.72
Few-shot (Brown et al., 2020)	74.88 $\pm$ 0.60	74.12 $\pm$ 1.88	74.35 $\pm$ 0.97	14.35 $\pm$ 0.39	10.20 $\pm$ 0.34	10.80 $\pm$ 2.07	23.10 $\pm$ 2.12	2.45 $\pm$ 1.06	13.90 $\pm$ 0.47
APO (Pryzant et al., 2023)	77.25 $\pm$ 1.66	76.90 $\pm$ 2.05	76.58 $\pm$ 0.30	24.10 $\pm$ 2.32	20.55 $\pm$ 2.23	18.40 $\pm$ 1.23	27.55 $\pm$ 1.53	14.20 $\pm$ 2.31	19.80 $\pm$ 1.76
GradeOpt (Chu et al., 2025)	82.94 $\pm$ 0.49	83.12 $\pm$ 2.15	82.85 $\pm$ 0.27	21.80 $\pm$ 1.22	15.40 $\pm$ 1.85	14.95 $\pm$ 1.15	35.20 $\pm$ 1.80	14.85 $\pm$ 1.63	25.60 $\pm$ 1.84
GEPA (Agrawal et al., 2026)	75.60 $\pm$ 1.89	74.90 $\pm$ 0.85	75.10 $\pm$ 1.48	21.30 $\pm$ 0.82	17.65 $\pm$ 1.28	15.20 $\pm$ 1.79	21.40 $\pm$ 0.25	10.30 $\pm$ 0.35	14.15 $\pm$ 0.63
RUBRICWORLD	81.15 $\pm$ 0.97	81.38 $\pm$ 0.36	81.62 $\pm$ 2.33	46.70 $\pm$ 1.57	45.10 $\pm$ 1.61	43.25 $\pm$ 0.88	39.15 $\pm$ 0.85	19.85 $\pm$ 1.63	29.70 $\pm$ 2.27

Table 1: **Final grading scores across three benchmarks and frozen LLM graders.** Three-seed mean $\pm$ SD; ACC/F1 in %, QWK on a 0–100 scale. Blue/red denotes changes from zero-shot; bold marks the best displayed score.



Figure 2: **Prediction, selection, and certification on ASAP2.0 (gpt-5-mini).** (a) Held-out-state MAE (pp); (b) selected-edit correction and degradation (%); (c) test QWK (0–100) and degradation (%). Three-seed mean $\pm$ SD; variants in Appendix O.

Variant	QWK@5 $\uparrow$	ACC@5 $\uparrow$	Tokens (M) $\downarrow$	
			5 rounds	20 rounds
Full RubricWorld	38.43	40.18	6.84	8.32
w/o State	33.16	35.42	6.41	8.68
w/o Learned Model	24.85	27.31	6.72	8.15

Table 2: **Edit-effect model ablation on ASAP2.0.** Average across three frozen graders with shared generator and certification.

frozen grader. Other feedback-driven prompt optimizers include OPRO and TextGrad (Yang et al., 2024; Yuksekgonul et al., 2025), which are dis-

cussed as related approaches but are not additional baselines in Table 1.

Editing and evaluation. The main comparison uses five refinement rounds and three random seeds. RUBRICWORLD considers five candidate edits per round. Candidate construction and paired-effect learning use $\mathcal{D}_{\text{dyn},\text{train}}$; $\mathcal{D}_{\text{dyn},\text{val}}$ is reserved for model tuning, $\mathcal{D}_{\text{cert}}$ for commit decisions, and $\mathcal{D}_{\text{test}}$ for final evaluation. The same answer is scored before and after an edit to measure what was corrected and what was newly broken; the test set is not used to choose edits. Baselines use their

native search procedures within the declared refinement protocol; implementation details are in Appendix D.

What we measure. We report exact-score accuracy (ACC), quadratic weighted kappa (QWK), and macro-F1 for final grading quality (Cohen, 1968; Attali and Burstein, 2006). For edit-effect prediction, we report errors in the predicted correction and degradation rates (MAE_C and MAE_D) in percentage points (pp); comparisons use unseen rubric states rather than random transition splits. For selection, we measure the chosen edit’s actual target correction, mean degradation, and empirical feasibility rate (EFR). For commit policies, we compare final QWK and observed degradation under different acceptance rules. Metric definitions appear in Appendix N.

Cost and long-horizon protocol. Optimization cost counts LLM tokens spent on proposing edits, collecting warm-start and online observations, paired scoring, and independent commit checks, excluding the shared final evaluation. The cost sweep reports three-seed mean QWK versus optimization tokens for gpt-5-mini.

The 20-round continuation retains the five-round settings. The 100-round plots use matched ASAP2.0 / gpt-5-mini runs, tracking validation ACC and residual degradation separately.

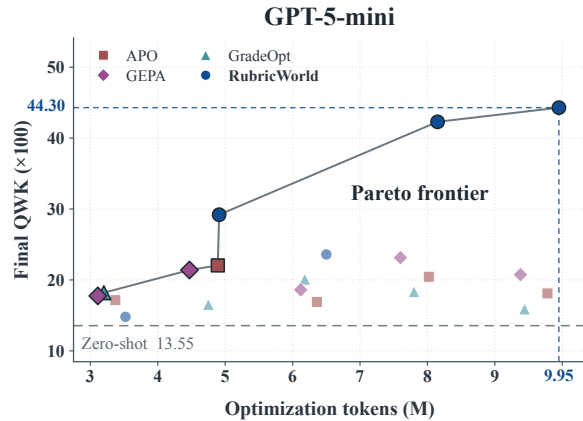


Figure 3: **Quality-cost trade-off on ASAP2.0 (gpt-5-mini).** Three-seed means including warm-start tokens; outlined points are Pareto-optimal.

5 Results and Analysis

5.1 Overall Grading Quality

RubricWorld’s strongest gains appear on ASAP2.0, while the short-answer results are less uniform. Table 1 compares three datasets and

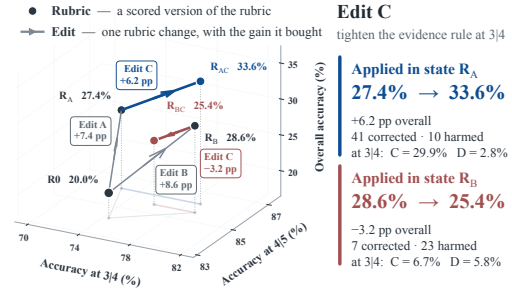


Figure 4: **Illustrative rubric-editing trajectories (ASAP2.0 scale; gpt-5-mini setting).** The same edit C is applied after A or B.

three frozen LLM graders. On ASAP2.0, RUBRIC-WORLD obtains QWK values of 32.94, 37.25, and 45.10 with Llama-3.1-8B, QwQ-32B, and gpt-5-mini; the strongest respective baselines reach 8.92, 28.15, and 20.55. All nine ASAP2.0 metric comparisons favor RUBRICWORLD.

Accuracy and ordinal agreement can disagree.

On EIR with QwQ-32B, APO beats RUBRIC-WORLD in ACC (75.30% versus 73.85%) but trails in QWK (71.85 versus 74.12), which penalizes larger scoring errors.

The advantage does not extend to every dataset and metric. On EIR with gpt-5-mini, GradeOpt slightly exceeds RUBRICWORLD in ACC and QWK; on ASAP-SAS with Llama-3.1-8B, GradeOpt has higher ACC. These exceptions limit the claim: gains are strongest and most consistent on ASAP2.0, not uniform across all grading settings.

5.2 Grading Quality Versus Token Cost

The quality gap between optimizers changes with the available token budget. Figure 3 compares final QWK on ASAP2.0 under the same frozen gpt-5-mini grader. In the budget sweep, RUBRICWORLD reaches 44.30 QWK near 10M tokens; at smaller budgets the methods are closer. The budget sweep reports token-constrained checkpoints, whereas Table 1 reports a fixed five-round endpoint (45.10 QWK); these are not identical checkpoints.

The additional fifteen rounds preserve the same settings, yet the methods accumulate different token costs. Table 2 gives cumulative tokens of 6.84M, 6.41M, and 6.72M after five rounds for the full, state-free, and model-free variants. At twenty rounds the totals are 8.32M, 8.68M, and 8.15M under the unchanged per-round protocol. The full model then uses fewer tokens than the state-free

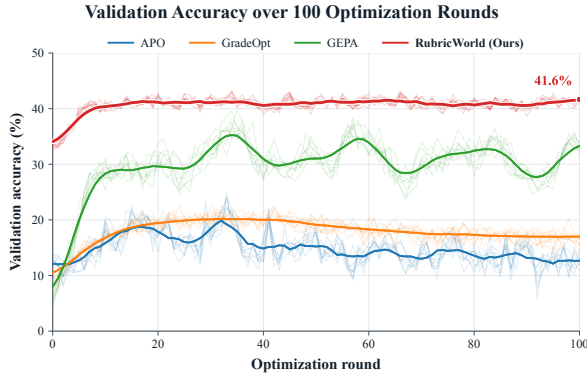


Figure 5: 100-round validation ACC.

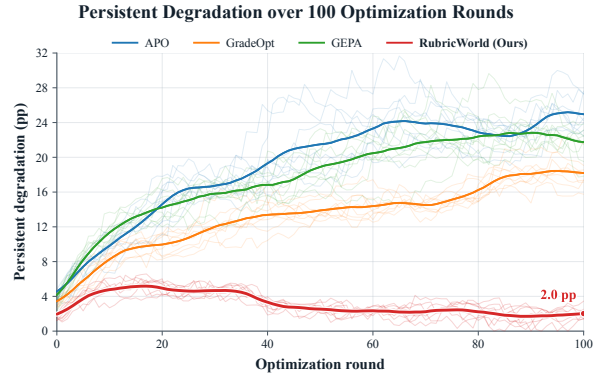


Figure 6: 100-round persistent degradation (pp).

variant, but more than direct evaluation. The totals do not identify which calls caused the difference; QWK and ACC concern only five rounds.

5.3 Why the Current Rubric Matters

The same edit can have opposite effects when its starting rubric changes, even at similar initial accuracy. Figure 4 illustrates Edit C after either A or B. The starting accuracies are 27.4% and 28.6%, but C raises the first to 33.6% and lowers the second to 25.4%: changes of +6.2 and −3.2 percentage points for the identical edit.

Different corrections and degradations explain the direction of each trajectory. On the A branch, C corrects 41 answers and harms 10; on the B branch, it corrects 7 and harms 23. At score boundary 3|4, the respective correction/degradation rates are 29.9%/2.8% and 6.7%/5.8%. A candidate’s text alone therefore does not determine the outcome without considering the current rubric.

The illustrative trajectory motivates state conditioning, but does not establish the prevalence of this effect or a predictor’s accuracy. Its axes are grading-accuracy coordinates, not learned state representations. Predictive value must instead be assessed on unseen rubric states and through the edits actually selected.

5.4 Prediction, Selection, and Commit Decisions

State conditioning improves prediction and selection. Figure 2(a,b) shows that the full model lowers MAE_C from 39.1 to 24.6 pp and MAE_D from 34.5 to 19.8 pp versus edit-only prediction (additive history ties on MAE_D). Its selected edits raise target correction from 8.5% to 16.2% and lower degradation from 8.8% to 2.5%.

Learned effects improve five-round grading. Ta-

ble 2 gives ACC/QWK of 40.18%/38.43 (full), 35.42%/33.16 (no state), and 27.31%/24.85 (direct candidate evaluation without learned prediction; Appendix O). Matched-seed means do not establish significance for the smaller QWK gap.

Certification trades 0.9 QWK for 4.6 pp less test degradation. In Figure 2(c), QWK/degradation is 46.0/7.8% (empirical gate), 45.1/3.2% (certification), and 41.2/14.2% (prediction-only); these are test measurements, not certification bounds.

5.5 Grading After 100 Refinement Rounds

Final grading quality and regressions left by earlier edits follow different trajectories. Figures 5 and 6 follow the same three 100-round ASAP2.0 runs with the frozen gpt-5-mini grader. RUBRICWORLD stays near 41% validation ACC while persistent degradation falls toward two percentage points.

A later revision may remove earlier damage without producing a corresponding jump in overall accuracy. An old regression leaves the persistence measure only after a later edit repairs that boundary decision. Other optimizers can recover aggregate accuracy while retaining more of the earlier damage. Unlike one edit’s conditional degradation rate, this metric counts unrepaired past errors. No joint finite-sample guarantee is claimed.

6 Conclusion

RUBRICWORLD learns state-dependent rubric-edit effects to guide candidate selection, then uses independent held-out certification to decide which edits to commit. Paired measurements update the predictor even when an edit is rejected. Across three educational grading benchmarks, the strongest gains occur on ASAP2.0; long-horizon results track persistent grading regressions alongside accuracy.

Limitations

RUBRICWORLD relies on initial rubrics and human reference scores; high reference agreement does not inherently guarantee educational validity when human targets contain noise (Messick, 1995). Empirical certification requires enough examples at checked boundaries; rare score categories may prevent otherwise valid commits. The edit-effect model may not generalize to unfamiliar criteria or tasks. Our evaluation covers educational grading, not open-domain judging, safety evaluation, or temporal streaming responses (Hashemi et al., 2024).

References

- Lakshya A Agrawal, Shangyin Tan, Dilara Soylu, Noah Ziemis, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J Ryan, Meng Jiang, Christopher Potts, Koushik Sen, Alex Dimakis, Ion Stoica, Dan Klein, Matei Zaharia, and Omar Khattab. 2026. *GEPA: Reflective prompt evolution can outperform reinforcement learning*. In *International Conference on Learning Representations*.
- Rishab Alagharu, Hongji Pu, Zeeshan Memon, Xinyuan Song, Yuntong Hu, and Liang Zhao. 2026. *Network World Models as Environments for Algorithm Design on Complex Systems*. *Preprint*, arXiv:2610.01048.
- Yigal Attali and Jill Burstein. 2006. *Automated essay scoring with e-rater v.2*. *The Journal of Technology, Learning and Assessment*, 4(3).
- Barbara, Ben Hamner, Jaison Morgan, lynnvande, and Mark Shermis. 2012. *The hewlett foundation: Short answer scoring*. Kaggle Competition.
- Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, and 12 others. 2020. *Language models are few-shot learners*. In *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901.
- Steven Burrows, Iryna Gurevych, and Benno Stein. 2015. *The eras and trends of automatic short answer grading*. *International Journal of Artificial Intelligence in Education*, 25(1):60–117.
- Wenzhi Cao, Vahid Mirjalili, and Sebastian Raschka. 2020. *Rank consistent ordinal regression for neural networks with application to age estimation*. *Pattern Recognition Letters*, 140:325–331.
- Rujikorn Charakorn, Edoardo Cetin, Shinnosuke Uesaka, and Robert Tjarko Lange. 2026. *Doc-to-LoRA: Learning to instantly internalize contexts*. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 12967–12997. PMLR.
- Guiming Hardy Chen, Shunian Chen, Ziche Liu, Feng Jiang, and Benyou Wang. 2024. *Humans or LLMs as the judge? a study on judgement bias*. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 8301–8327, Miami, Florida, USA. Association for Computational Linguistics.
- Yucheng Chu, Hang Li, Kaiqi Yang, Yasemin Copur-Gencurk, Joseph Krajcik, Namsoo Shin, and Jiliang Tang. 2026. *Confusion-aware rubric optimization for LLM-based automated grading*. In *Proceedings of the 19th International Conference on Educational Data Mining*, pages 67–78, Seoul, Republic of Korea. International Educational Data Mining Society.
- Yucheng Chu, Hang Li, Kaiqi Yang, Harry Shomer, Yasemin Copur-Gencurk, Leonora Kaldaras, Kevin Haudek, Joseph Krajcik, Namsoo Shin, Hui Liu, and Jiliang Tang. 2025. *A LLM-powered automatic grading framework with human-level guidelines optimization*. In *Proceedings of the 18th International Conference on Educational Data Mining*, pages 31–41, Palermo, Italy. International Educational Data Mining Society.
- C. J. Clopper and E. S. Pearson. 1934. *The use of confidence or fiducial limits illustrated in the case of the binomial*. *Biometrika*, 26(4):404–413.
- Jacob Cohen. 1968. *Weighted kappa: Nominal scale agreement with provision for scaled disagreement or partial credit*. *Psychological Bulletin*, 70(4):213–220.
- Scott A. Crossley, Perpetual Baffour, L. Burleigh, and Jules King. 2025. *A large-scale corpus for assessing source-based writing quality: ASAP 2.0*. *Assessing Writing*, 65:100954.
- Scott A. Crossley, Yu Tian, Perpetual Baffour, Alex Franklin, Meg Benner, and Ulrich Boser. 2024. *A large-scale corpus for assessing written argumentation: PERSUADE 2.0*. *Assessing Writing*, 61:100865.
- Florian Eddie Dorner, Vivian Yvonne Nastl, and Moritz Hardt. 2025. *Limits to scalable evaluation at the frontier: LLM as judge won’t beat twice the data*. In *International Conference on Learning Representations*.
- Cynthia Dwork, Vitaly Feldman, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Aaron Roth. 2015. *The reusable holdout: Preserving validity in adaptive data analysis*. *Science*, 349(6248):636–638.
- Diana Galvan-Sosa, Gabrielle Gaudeau, Pride Kavumba, Yunmeng Li, Hongyi Gu, Zheng Yuan, Keisuke Sakaguchi, and Paula Buttery. 2025. *Rubrik’s cube*.

794	score meaning . <i>American Psychologist</i> , 50(9):741–	Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane	848
795	749.	Suhr. 2024. Quantifying language models’ sensitiv-	849
796	Michael Mohler, Razvan Bunescu, and Rada Mihalcea.	ity to spurious features in prompt design or: How i	850
797	2011. Learning to grade short answer questions us-	learned to start worrying about prompt formatting.	851
798	ing semantic similarity measures and dependency	In <i>International Conference on Learning Representa-</i>	852
799	graph alignments . In <i>Proceedings of the 49th An-</i>	<i>tions</i> .	853
800	<i>Annual Meeting of the Association for Computational</i>	Mark D. Shermis and Ben Hamner. 2013. Contrast-	854
801	<i>Linguistics: Human Language Technologies</i> , pages	ing state-of-the-art automated scoring of essays . In	855
802	752–762, Portland, Oregon, USA. Association for	Mark D. Shermis and Jill Burstein, editors, <i>Hand-</i>	856
803	Computational Linguistics.	<i>book of Automated Essay Evaluation: Current Appli-</i>	857
804	OpenAI. 2026. GPT-5 mini model . OpenAI API Doc-	<i>cations and New Directions</i> , pages 313–346. Rout-	858
805	umentation. Undated official OpenAI API model	ledge.	859
806	documentation; accessed 2026-10-08. Year denotes	Kaveh Taghipour and Hwee Tou Ng. 2016. A neural	860
807	access year, not publication date.	approach to automated essay scoring . In <i>Proceedings</i>	861
808	Krista Opsahl-Ong, Michael J Ryan, Josh Purtell, David	<i>of the 2016 Conference on Empirical Methods in Nat-</i>	862
809	Broman, Christopher Potts, Matei Zaharia, and Omar	<i>ural Language Processing</i> , pages 1882–1891, Austin,	863
810	Khattab. 2024. Optimizing instructions and demon-	Texas. Association for Computational Linguistics.	864
811	strations for multi-stage language model programs .	The Learning Agency Lab. 2024. Learning agency lab -	865
812	In <i>Proceedings of the 2024 Conference on Empiri-</i>	automated essay scoring 2.0 . Kaggle Competition.	866
813	<i>cal Methods in Natural Language Processing</i> , pages	Tu Vu, Kalpesh Krishna, Salaheddin Alzubi, Chris Tar,	867
814	9340–9366, Miami, Florida, USA. Association for	Manaal Faruqui, and Yun-Hsuan Sung. 2024. Foun-	868
815	Computational Linguistics.	dational autoraters: Taming large language models	869
816	Ellis B. Page. 1966. The imminence of... grading essays	for better automatic evaluation . In <i>Proceedings of the</i>	870
817	by computer . <i>Phi Delta Kappan</i> , 47(5):238–243.	<i>2024 Conference on Empirical Methods in Natural</i>	871
818	Reid Pryzant, Dan Iter, Jerry Li, Yin Lee, Chenguang	<i>Language Processing</i> , pages 17086–17105, Miami,	872
819	Zhu, and Michael Zeng. 2023. Automatic prompt op-	Florida, USA. Association for Computational Lin-	873
820	timization with “gradient descent” and beam search .	guistics.	874
821	In <i>Proceedings of the 2023 Conference on Empiri-</i>	Xinyuan Wang, Chenxi Li, Zhen Wang, Fan Bai, Hao-	875
822	<i>cal Methods in Natural Language Processing</i> , pages	tian Luo, Jiayou Zhang, Nebojsa Jojic, Eric P. Xing,	876
823	7957–7968, Singapore. Association for Computa-	and Zhiting Hu. 2024. PromptAgent: Strategic	877
824	tional Linguistics.	planning with language models enables expert-level	878
825	Hongji Pu, Ruixiang Tang, and Yongfeng Zhang. 2026.	prompt optimization . In <i>International Conference on</i>	879
826	COUNTERMEM: World-model verified counter-	<i>Learning Representations</i> .	880
827	factual memory for language agents . <i>Preprint</i> ,	Yifan Xu, Yixuan Li, Xinzhuo Li, Yixin Gu, Yifan Shen,	881
828	arXiv:2609.31874.	Lijun Yu, and Haohan Wang. 2026. STEVE: Stabi-	882
829	Martin L. Puterman. 1994. <i>Markov Decision Processes:</i>	lizing textual gradient-based prompt optimization via	883
830	<i>Discrete Stochastic Dynamic Programming</i> . Wiley.	error-driven refinement and regularized verification .	884
831	Qwen Team. 2024. QwQ-32b-preview . Hugging Face	<i>Preprint</i> , arXiv:2609.23716.	885
832	Model Card.	Chengrun Yang, Xuezhi Wang, Yifeng Lu, Hanxiao Liu,	886
833	Steffen Rendle. 2010. Factorization machines . In	Quoc V. Le, Denny Zhou, and Xinyun Chen. 2024.	887
834	<i>2010 IEEE International Conference on Data Mining</i> ,	Large language models as optimizers . In <i>Interna-</i>	888
835	pages 995–1000.	<i>tional Conference on Learning Representations</i> .	889
836	Keisuke Sakaguchi, Michael Heilman, and Nitin Mad-	Mert Yuksekogonul, Federico Bianchi, Joseph Boen,	890
837	nani. 2015. Effective feature integration for auto-	Sheng Liu, Pan Lu, Zhi Huang, Carlos Guestrin,	891
838	mated short answer scoring . In <i>Proceedings of the</i>	and James Zou. 2025. Optimizing generative AI by	892
839	<i>2015 Conference of the North American Chapter of</i>	backpropagating language model feedback . <i>Nature</i> ,	893
840	<i>the Association for Computational Linguistics: Hu-</i>	639(8055):609–616.	894
841	<i>man Language Technologies</i> , pages 1049–1054, Den-	Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan	895
842	ver, Colorado. Association for Computational Lin-	Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin,	896
843	guistics.	Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang,	897
844	Subhash Bangalore Satheesha, Nirvik Pande, Deepthi	Joseph E. Gonzalez, and Ion Stoica. 2023. Judging	898
845	Duddempudi, and Bharath Dandala. 2026. RLMOpt:	llm-as-a-judge with mt-bench and chatbot arena . In	899
846	Adaptive prompt optimization via recursive language	<i>Advances in Neural Information Processing Systems</i> ,	900
847	models . <i>Preprint</i> , arXiv:2608.10471.	volume 36, pages 46595–46623. Curran Associates,	901
		Inc.	902

Jin Peng Zhou, Séb Arnold, Nan Ding, Kilian Q Wein-
berger, Nan Hua, and Fei Sha. 2025. [Graders should
cheat: Privileged information enables expert-level au-
tomated evaluations](#). In *Proceedings of the 2025 Con-
ference on Empirical Methods in Natural Language
Processing*, pages 16572–16590, Suzhou, China. As-
sociation for Computational Linguistics.

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B Dataset Counts and Data Use

Our essay tasks assign each response one ordered score, as in earlier automated essay scoring comparisons (Shermis and Hamner, 2013). Short-answer tasks use question-specific scoring criteria (Burrows et al., 2015). The configuration specifies stratification by reference score when validation responses are selected from the training data. Unless otherwise stated, optimization uses only the training split and validation subsets; the final test set is held out until the final evaluation.

B.1 EIR (National)

EIR (National) contains five private short-answer items, each with an item-specific score scale. To document the experimental split without exposing restricted response content, Table 3 reports aggregate counts only.

Item	<i>n</i>	Scale	Train	Val/Test
U41Q1	199	0–3	159	20 / 20
U41Q2	230	0–4	184	23 / 23
U42Q1	248	0–4	198	25 / 25
U42Q2	262	0–3	209	26 / 27
U43Q1	434	0–4	347	43 / 44
Total	1,373	–	1,097	137 / 139

Table 3: EIR (National) aggregate split statistics. Items use fixed 80/10/10 partitions, stratified by reference score where class counts permit.

No response appears in more than one partition, and the test partition is reserved for final evaluation, consistent with the protocol in Section 4.

B.2 ASAP2.0

ASAP2.0 uses a 1–6 holistic scoring scale (The Learning Agency Lab, 2024). Its essays come from the PERSUADE 2.0 corpus (Crossley et al., 2024). The task requires evaluating student essays according to argument quality, evidence use, organization, language control, and mechanics. Table 4 reports the split statistics used in our experiments.

Split	#Samples	Score distribution
Train	14,711	1058 / 4023 / 5326 / 3350 / 827 / 127
Test	2,596	194 / 700 / 954 / 576 / 143 / 29

Table 4: ASAP2.0 split statistics. Score distributions are ordered from score 1 to score 6.

B.3 ASAP-SAS

ASAP-SAS consists of item-specific short-answer scoring tasks (Barbara et al., 2012). We use the public train/test split and carve out a validation set from the training data. The score scale is item-specific; for the reported Set 1 experiments, the score range is 0–3. We use Set 1 because its four score levels are sufficiently populated to support the three boundary-level diagnostics; we do not claim that this result transfers to the remaining items. Table 5 lists the training, validation, and test counts.

Split	#Samples	Score distribution
Train	1,170	266 / 301 / 366 / 237
Valid	252	Stratified from training data
Test	251	57 / 64 / 79 / 51

Table 5: ASAP-SAS Set 1 split statistics. Score distributions are ordered from score 0 to score 3.

B.4 Data Access: Private and Public Datasets

EIR (National) is a private dataset and is not publicly available. It contains written responses used to assess teachers’ subject knowledge and knowledge of teaching (Chu et al., 2025). The other two datasets, ASAP2.0 and ASAP-SAS, are public benchmarks. Their availability does not imply unrestricted reuse or redistribution; users should follow the original providers’ access and usage terms.

C Prompt and Patch Format

The scoring prompt is the text sent to the frozen grader. A patch proposes an edit to that text and records why the change is needed. The formats below show the initial essay rubric, the information supplied with a candidate edit, and diagnostic paired-effect records; independent certification makes the final commit decision (Appendix L).

C.1 Initial Scoring Prompt

The initial prompt tells the grader what each essay score means. It is the starting specification; optimization adds or revises scoring instructions with-

out changing the grader’s parameters. This differs from approaches that train rubric-following evaluators or construct explanatory evaluation rubrics (Kim et al., 2024b; Galvan-Sosa et al., 2025).

Initial Holistic Scoring Prompt

You are grading a student essay response on a 1-6 holistic scale.

Rubric dimensions:

- Point of view and critical thinking
- Use of examples, reasons, and evidence
- Organization, focus, coherence, and progression
- Language quality, including vocabulary and sentence structure
- Grammar, usage, and mechanics

Score 6:

Clear and consistent mastery. The essay presents a strong position, well-developed reasoning, relevant evidence, coherent organization, and skillful language control.

Score 5:

Reasonably consistent mastery with occasional lapses.

Score 4:

Adequate mastery with lapses in quality.

Score 3:

Developing mastery. The essay shows partial control of the task but has limited support, weak explanation, inconsistent organization, or accumulating language problems.

Score 2:

Little mastery. The essay may contain a vague position or minimal support, with weak organization and frequent language problems.

Score 1:

Very little or no mastery. The essay is highly limited, disorganized, off-topic, or difficult to understand.

C.2 Boundary-Targeted Patch Schema

Each candidate contains the proposed scoring text and the evidence for it. In the schema, `patch_text` is the edit applied to the specification; the other fields identify its target and explain the intended effect.

Boundary-Targeted Patch Schema

```
{
  "target_boundary": "<score
    transition b|b+1>",
  "error_direction": "<over-scoring |
    under-scoring>",
  "criterion": "<affected rubric
    criterion>",
  "failure_cases": "<misgraded
    examples requiring correction>",
  "anchor_cases": "<correctly scored
    training examples>",
  "patch_text": "<proposed edit to the
    scoring instructions>",
  "expected_effect": "<intended
    boundary-level correction>",
  "rationale": "<why this update
    addresses the observed failure>"
}
```

A patch for 3 | 4 should clarify when a response deserves at least score 4. Failure cases illustrate mistakes to repair, while training anchors are correctly scored examples supplied to the generator. The intended scope guides candidate filtering, while paired-effect auditing measures the resulting grading changes.

C.3 Patch Generation Constraints

The generation instructions ask for an edit that addresses the observed failure, retains the original criteria, and targets the selected score distinction. These requests do not by themselves guarantee compliance. For example, an edit about explaining evidence should not add a new requirement unrelated to the rubric.

We represent the candidate as $p = (b, d_{\text{dir}}, \mathcal{F}^{\text{train}}, \mathcal{A}^{\text{train}}, \Delta R)$: b is its intended boundary, d_{dir} is over- or under-scoring, $\mathcal{F}^{\text{train}}$ contains training failures, $\mathcal{A}^{\text{train}}$ contains training anchors, and ΔR is the proposed text edit. The target metadata identifies intended corrections; paired grading measures the effects of the proposed edit.

C.4 Audit Metadata

A selected candidate is associated with diagnostic paired-effect measurements and a separate held-out certification outcome:

Patch Audit Record

```
{
  "target_correction": "<paired target
    -boundary correction>",
```

Method	Text supplied or changed	Selection basis
Zero-shot	Original rubric	No update
Few-shot	Scored examples appended to prompt	Validation agreement
APO	Rewritten scoring instructions	Aggregate validation score
GradeOpt	Guidelines and examples derived from errors	Validation improvement
GEPA	Prompt variants generated from reflection	Candidate-pool selection
RUBRICWORLD	Edit to a chosen score distinction	Repair and per-boundary breakage

Table 6: What each comparison method changes and evaluates.

```

"boundary_degradation": "<paired
degradation by boundary>",
"repair_support": "<eligible paired
counts by boundary>",
"degradation_support": "<eligible
paired counts by boundary>",
"certification_repair_support": "<
held-out target-error count>",
"certification_degradation_support":
"<held-out initially-correct
counts by boundary>",
"certification_status": "<certified
| rejected | unsupported>",
"commit_status": "<accepted |
retained>"
}

```

The paired edit-effect record contains correction and degradation rates at the observed score boundaries. These measurements update the learned predictor irrespective of the subsequent commit decision. The final acceptance rule instead applies exact one-sided confidence bounds on the unused certification split: a candidate is committed only when the target correction lower bound and every degradation upper bound in the prespecified certification scope meet their respective thresholds (Section 3.5; Appendix L.3). Residual boundary regressions are tracked separately for long-horizon analysis (Appendix I.6).

D Baseline Implementation Details

The methods differ in what they add to or revise in the scoring specification, and how they choose the resulting text. The grading LLM is frozen for every method.

APO uses feedback on grading errors to propose revised instructions, then selects by aggregate val-

Parameter	Value
Optimization rounds	5
Main-table repetitions	3 seeds; mean $\pm$ std
Long-horizon analysis rounds	100
Candidate patches per round	5
Validation sampling	By reference score
Data used to find errors	Training partition
Minimum target repair γ	0.80
Protected regression tolerance ϵ_b	0.08
Intended patch scope	One score boundary

Table 7: Reported optimization settings for RUBRICWORLD.

idation performance. GradeOpt turns misgraded responses into revised guidelines and examples. GEPA uses reflection on evaluation traces to generate prompt variants and maintains a pool of candidates. RUBRICWORLD evaluates an edit by the intended failures it repairs and the previously correct decisions it breaks at each score boundary. Their search strategies follow the original method descriptions, while the frozen grader is shared across comparisons.

E Hyperparameter Settings

Table 7 lists the reported settings. A round is one cycle of target selection, candidate generation, and acceptance checking. A patch count refers to proposed text edits, including rejected candidates.

F Audit Threshold Selection

The two thresholds constrain different one-sided confidence bounds. $\gamma = 0.80$ is the minimum required lower bound for target-boundary correction; $\epsilon_b = 0.08$ is the maximum allowed upper bound for degradation on each boundary in the prespecified certification scope. For illustration, a sample with 20 target failures and 100 initially correct decisions would have empirical point estimates of 80% correction and 8% degradation after 16 repairs and eight regressions. Certification, however, evaluates the corresponding confidence limits rather than these point estimates (Appendix L.2).

The thresholds are fixed before final test evaluation and shared across datasets and graders; improvements at one boundary do not offset a bound violation at another certified boundary.

G Ethical Considerations

RUBRICWORLD supports research on educational scoring; passing its audit does not establish suit-

ability for high-stakes grading. Incorrect scores can affect students’ opportunities, so consequential or disputed decisions should receive human review, with a way to contest errors. Agreement with reference scores does not establish fairness: references may contain bias, and average metrics can hide unequal errors across groups. Use in practice would require checking these differences in the intended population. Student and teacher responses may also contain identifying information. Sharing them, including with external model services, should respect access conditions and minimize disclosure of personal information. These obligations apply to both the private EIR (National) dataset and the public ASAP2.0 and ASAP-SAS benchmarks (Appendix B.4).

H Data Isolation and Candidate Construction

This section provides the data-isolation protocol and candidate construction procedure used by the optimization loop in Section 3.

H.1 Data Isolation

We split labeled answers into four disjoint sets:

$$\mathcal{D}_{\text{dyn,train}}, \mathcal{D}_{\text{dyn,val}}, \mathcal{D}_{\text{cert}}, \mathcal{D}_{\text{test}}.$$

Training responses supply candidate construction, paired-effect measurements, and model fitting. The dynamics-validation set is reserved for predictor tuning; the certification set governs commit decisions and the final test set is accessed only for final evaluation.

Grouped prediction evaluation is a separate procedure. Within the dynamics-training data, we separately form held-out folds by `edit_id` (edit-instance groups), `edit_text` (unseen clauses), and `rubric_state_id` (unseen states). Every example from a held-out group stays outside that fold’s model training data; group-level results are not used for certification. This grouping tests prediction generalization and does not replace the four dataset partitions or the final test holdout.

Partition membership is fixed before optimization. Responses used for response-model fitting or candidate construction are never reused as certification examples. Likewise, no information from $\mathcal{D}_{\text{test}}$ is used for candidate generation, response prediction, candidate selection, or certification.

When duplicate response texts occur in the original data, duplicates are assigned to the same partition before optimization. This prevents identical

responses from entering both the model-training and certification pathways.

H.2 Round-Wise Certification Splits

For a run containing T optimization rounds, the certification data are partitioned in advance into disjoint subsets:

$$\mathcal{D}_{\text{cert}} = \bigcup_{t=0}^{T-1} \mathcal{D}_{\text{cert}}^{(t)},$$

with

$$\mathcal{D}_{\text{cert}}^{(t)} \cap \mathcal{D}_{\text{cert}}^{(t')} = \emptyset \quad \text{for } t \neq t'.$$

Round t accesses only $\mathcal{D}_{\text{cert}}^{(t)}$. Therefore, certification outcomes from earlier rounds cannot reveal examples that will later be used to certify another update.

This distinction is important because candidate generation and response prediction adapt to earlier optimization outcomes. Fresh round-specific certification subsets keep the final commit decision separate from this adaptive optimization process. The outer benchmark split counts in Appendix B are distinct from the eligible before-edit correction and degradation counts used in each certificate. Those round-level denominators are n_t^C and $n_{t,b}^D$ in Appendix L.1.

H.3 Certification Scope

For contiguous score labels from $y_{\min}$ to $y_{\max}$, let

$$\mathcal{B} = \{y_{\min}, \dots, y_{\max} - 1\}$$

denote the actual score boundaries, including 0 | 1 on a 0–3 item. Finite-sample guarantees are reported only over a pre-specified subset

$$\mathcal{B}_{\text{cert}} \subseteq \mathcal{B}.$$

The certification scope $\mathcal{B}_{\text{cert}}$ is fixed before candidate outcomes are inspected. Each required confidence bound is then checked using the corresponding eligible certification decisions.

Boundaries outside $\mathcal{B}_{\text{cert}}$ remain part of the edit response and are still considered by the response model and candidate selection rule. They are reported empirically, but no finite-sample certification claim is made for them.

1162 H.4 Error Groups

1163 At optimization round t , the frozen grader evaluates
 1164 $\mathcal{D}_{\text{dyn,train}}$ using the current scoring specification
 1165 R_t . Each incorrect prediction is associated with the
 1166 score boundaries that separate the predicted score
 1167 from its reference score.

1168 Candidate construction operates on localized error
 1169 groups. We write a group as

$$1170 g = (b, d_{\text{dir}}, o),$$

1171 where b denotes the affected score boundary, d_{dir} is
 1172 the over- or under-scoring direction, and o records
 1173 whether the error was already present or appeared
 1174 after an earlier accepted revision.

1175 The origin field is used to describe the observed
 1176 optimization history; it is not treated as a causal
 1177 explanation of the error.

1178 Let $\mathcal{E}_t$ be the set of training-answer grading errors
 1179 at round t , and let $\mathcal{G}_t$ be the groups formed
 1180 from those errors. The routing rule selects

$$1181 g_t = \rho_t(\mathcal{E}_t) = (b_t^*, d_{t,\text{dir}}, o_t) \in \mathcal{G}_t$$

1182 for candidate construction. The corresponding
 1183 boundary b_t^* becomes the target boundary for that
 1184 optimization round.

1185 H.5 Scoped Candidate Generation

1186 The candidate generator receives the current scor-
 1187 ing specification R_t , examples from the selected error
 1188 group, and relevant correctly graded responses
 1189 from $\mathcal{D}_{\text{dyn,train}}$.

1190 The signed error direction first determines the in-
 1191 tended function of the revision. Depending on the
 1192 observed failure, the candidate may tighten a scor-
 1193 ing distinction, loosen it, or clarify an ambiguous
 1194 decision rule. The target and direction are deter-
 1195 mined before the final text of the candidate clause
 1196 is generated.

1197 The generator then produces a small collection

$$1198 \mathcal{P}_t = \{p_t^1, \dots, p_t^M\}$$

1199 of localized revisions. Each p_t^j describes a textual
 1200 modification to R_t rather than an entire replace-
 1201 ment rubric.

1202 Candidates are filtered before response predic-
 1203 tion. A candidate is removed if it merely duplicates
 1204 an existing rule, depends on the topic or wording
 1205 of a specific response, introduces an unrelated scor-
 1206 ing requirement, directly contradicts the current
 1207 specification, or duplicates another candidate in the
 1208 same pool.

Candidate generation itself never modifies the
 deployed scoring specification. For every candi-
 date,

$$R_t \oplus p_t^j$$

is a temporary specification until the candidate
 passes both model-guided selection and empirical
 certification.

I Edit-Response Measurement Details

This section expands the paired edit-response defi-
 nition introduced in Section 3.2.

I.1 Ordinal Boundary Representation

For contiguous labels $y_{\min}, \dots, y_{\max}$, $\mathcal{B} = \{y_{\min}, \dots, y_{\max} - 1\}$ contains all $K - 1$ thresholds. EIR items scored 0–3 use 0 | 1, 1 | 2, and 2 | 3; items scored 0–4 have four boundaries. Each score is represented by $K - 1$ binary decisions:

$$z_{i,b} = \mathbb{I}(y_i > b), \quad b \in \mathcal{B}.$$

Likewise, the prediction under the current rubric
 is

$$\hat{z}_{i,b}^{(t)} = \mathbb{I}(\hat{y}_i^{(t)} > b).$$

After applying candidate edit p ,

$$\hat{z}_{i,b}^p = \mathbb{I}(\hat{y}_i^p > b)$$

denotes the corresponding decision under $R_t \oplus p$.

A single score change can modify several bound-
 ary decisions. For example, moving from score 2
 to score 4 changes both the 2|3 and 3|4 decisions.
 The boundary representation therefore preserves
 information that would be lost if an edit were rep-
 resented only by the scalar difference between pre-
 dicted scores.

I.2 Repair and Regression Sets

Before applying candidate p , define

$$\mathcal{F}_{t,b} = \left\{ i : z_{i,b} \neq \hat{z}_{i,b}^{(t)} \right\},$$

and

$$\mathcal{A}_{t,b} = \left\{ i : z_{i,b} = \hat{z}_{i,b}^{(t)} \right\}.$$

$\mathcal{F}_{t,b}$ contains decisions that are currently wrong
 at boundary b . $\mathcal{A}_{t,b}$ contains decisions that are
 currently correct at the same boundary.

For nonempty $\mathcal{F}_{t,b}$, target repair is measured as

$$C_{t,b}(p) = \frac{1}{|\mathcal{F}_{t,b}|} \sum_{i \in \mathcal{F}_{t,b}} \mathbb{I}(z_{i,b} = \hat{z}_{i,b}^p).$$

For nonempty $\mathcal{A}_{t,b}$, regression is measured as

$$D_{t,b}(p) = \frac{1}{|\mathcal{A}_{t,b}|} \sum_{i \in \mathcal{A}_{t,b}} \mathbb{I}(z_{i,b} \neq \hat{z}_{i,b}^p).$$

Thus, $C_{t,b}$ measures errors repaired by the edit, while $D_{t,b}$ measures correct decisions damaged by the same edit.

I.3 Validity Indicators

Some response components can be undefined because their denominator is empty. We therefore define

$$m_{t,b}^C = \mathbb{I}(|\mathcal{F}_{t,b}| > 0),$$

and

$$m_{t,b}^D = \mathbb{I}(|\mathcal{A}_{t,b}| > 0).$$

The complete validity mask is

$$\mathbf{m}_t = [\mathbf{m}_t^C; \mathbf{m}_t^D].$$

Incomplete-record drop rule. We retain an answer–edit evaluation only if the reference, before-edit, and after-edit decisions are available at all $K - 1$ boundaries. If any boundary entry is missing, the entire record is dropped; no missing entry is padded or filled with zero. This differs from an empty correction/degradation denominator: such a well-formed transition is retained, but the undefined aggregate rate coordinate receives a zero mask and is excluded from the relevant prediction loss.

I.4 Response Vector

The response associated with candidate edit p is

$$\mathbf{e}_t(p) = [\mathbf{C}_t(p); \mathbf{D}_t(p)],$$

where

$$\mathbf{C}_t(p) = [C_{t,b}(p)]_{b \in \mathcal{B}},$$

and

$$\mathbf{D}_t(p) = [D_{t,b}(p)]_{b \in \mathcal{B}}.$$

The response dimension is therefore

$$d_e = 2(K - 1).$$

The coordinates follow the score boundaries in $\mathcal{B}$ for the scoring task under consideration. Thus $d_e = 10$ for a 1–6 scale, $d_e = 8$ for a 0–4 scale, and $d_e = 6$ for a 0–3 scale.

For a candidate targeting b_t^* ,

$$C_{t,b_t^*}(p)$$

is the intended repair coordinate. The remaining repair and regression coordinates describe effects away from the target boundary.

The boundary vector is therefore used as a coordinate system for the effect of an edit rather than as a separate optimization objective.

I.5 Transition Records

Whenever a selected candidate is evaluated on $\mathcal{D}_{\text{dyn},\text{train}}$, we record

$$(s_t, p_t^*, \mathbf{e}_t^{\text{obs}}, \mathbf{m}_t^{\text{obs}}).$$

This record is retained even when the candidate is later rejected by the certification stage. Certification controls whether the rubric changes; it does not alter the fact that the paired edit response has been observed.

The response history therefore evolves as

$$\mathcal{H}_{t+1} = \mathcal{H}_t \cup \left\{ (p_t^*, \mathbf{e}_t^{\text{obs}}, \mathbf{m}_t^{\text{obs}}) \right\}.$$

The supervised transition set becomes

$$\mathcal{T}_{t+1} = \mathcal{T}_t \cup \left\{ (s_t, p_t^*, \mathbf{e}_t^{\text{obs}}, \mathbf{m}_t^{\text{obs}}) \right\}.$$

Candidates that are never applied to $\mathcal{D}_{\text{dyn},\text{train}}$ have no observed transition and therefore do not enter $\mathcal{H}_t$ or $\mathcal{T}_t$.

I.6 Persistent Regressions

For trajectory-level analysis, we additionally track regressions that remain active after subsequent optimization rounds.

A decision becomes an active regression when an accepted edit changes a previously correct boundary decision into an incorrect one. It remains active until a later accepted edit restores that decision.

This diagnostic distinguishes transient damage from errors that persist through the optimization trajectory. It is used only for analysis and is not an additional term in the round-level optimization objective.

J State-Conditioned Edit-Effect Model Details

This section specifies the edit-effect model, its response targets, and the transition data used for supervised updating.

J.1 State Definition

The optimization state at round t is

$$s_t = (R_t, \mathcal{H}_t, \mathbf{v}_t),$$

where R_t is the active scoring rubric, $\mathcal{H}_t$ stores previously measured edit effects, and $\mathbf{v}_t$ summarizes the frozen grader’s behavior on labeled training answers. Only R_t conditions the frozen grader; $\mathcal{H}_t$ and $\mathbf{v}_t$ provide predictive information to the edit-effect model. The supervised transition set $\mathcal{T}_t$ is distinct from the state history $\mathcal{H}_t$.

For each training answer x_i , let $\pi_{i,k}^{(t)}$ denote its available score probability for label k . Define its cumulative probability above boundary b , threshold margin, and correctness as

$$q_{i,b}^{(t)} = \sum_{k>b} \pi_{i,k}^{(t)},$$

$$r_{i,b}^{(t)} = \left| q_{i,b}^{(t)} - \frac{1}{2} \right|, \quad c_{i,b}^{(t)} = \mathbb{I}(\hat{z}_{i,b}^{(t)} = z_{i,b}).$$

For a fixed scoring task and grader, we pool answer features and $\mathbf{c}_i$ into the fixed-dimensional summary $\mathbf{v}_t$, adding $\mathbf{q}_i$ and $\mathbf{r}_i$ only when the grader provides usable score probabilities. In particular, API-only graders need not expose per-label probabilities: missing probability and margin features are omitted, not inferred from token logits. All reference-based features use training labels only; neither certification nor test labels enter the state encoder.

J.2 Edit and State Encoders

The candidate-edit encoder is a deterministic mapping

$$\phi : p \mapsto a_p \in \mathbb{R}^{d_a}.$$

We consider three candidate-edit encodings:

$$\phi_{\text{text}}(p) = \text{Enc}(p, b^*, d_{\text{dir}}),$$

$$\phi_{\text{D2L}}(p) = \text{Pool}_{j=1}^J [\Delta W(p) u_j],$$

$$\phi_{\text{hybrid}}(p) = [\phi_{\text{text}}(p); \phi_{\text{D2L}}(p)].$$

The text encoder reads the candidate clause and target-boundary metadata. For the Doc-to-LoRA (D2L) sketch, the text-conditioned low-rank update has $\Delta W(p) = B_p A_p$ and fixed probe vectors u_j . Using $\Delta W(p) u_j$ rather than flattened LoRA factors avoids arbitrary factorization coordinates: for any invertible Q ,

$$(B_p Q)(Q^{-1} A_p) u_j = B_p A_p u_j.$$

Thus the probes are invariant to the LoRA gauge transformation. The hybrid encoder concatenates text and D2L features. These are alternative edit representations, not three simultaneous changes to the frozen grading model.

The state encoder $\psi(s_t)$ maps rubric text, historical edit measurements, and $\mathbf{v}_t$ into a fixed-dimensional vector $h_t \in \mathbb{R}^{d_h}$. The grader is never retrained. Grouped and edit-only comparisons hold the encoding protocol fixed unless the encoder itself is the factor being tested.

J.3 Factorized Edit–State Interaction

For a general state–edit pair (s, p) , let

$$a_p = \phi(p), \quad h_s = \psi(s).$$

The response logits are

$$\boldsymbol{\eta}_\theta(s, p) = W_a a_p + U [(A a_p) \odot (B h_s)] + c.$$

The predicted response is

$$\boldsymbol{\mu}_\theta(s, p) = \sigma(\boldsymbol{\eta}_\theta(s, p)),$$

where σ is applied element-wise.

The parameters have dimensions

$$W_a \in \mathbb{R}^{d_e \times d_a},$$

$$A \in \mathbb{R}^{r_{\text{int}} \times d_a}, \quad B \in \mathbb{R}^{r_{\text{int}} \times d_h},$$

$$U \in \mathbb{R}^{d_e \times r_{\text{int}}}, \quad c \in \mathbb{R}^{d_e}.$$

The term

$$W_a a_p$$

captures the component of the response associated with the candidate edit independently of the current state.

The interaction term

$$U [(A a_p) \odot (B h_s)]$$

captures how the same edit representation changes under different states.

The two projected vectors

$$A a_p, \quad B h_s \in \mathbb{R}^{r_{\text{int}}}$$

are combined through element-wise multiplication. The resulting interaction features are mapped by U into the repair–regression response space.

A full three-way interaction over edit, state, and response dimensions would require parameters proportional to $d_a d_h d_e$. The factorized form instead uses a shared interaction dimension r_{int} and avoids explicitly constructing that tensor.

J.4 Edit-Only Model

To test whether the current rubric state contributes predictive information beyond the candidate edit itself, we remove the edit–state interaction and use

$$\mu_{\text{edit}}(p) = \sigma(W_a a_p + c).$$

The response targets, edit representation, transition data, and training procedure remain unchanged. The difference between this model and the full model therefore isolates the value of conditioning on the current optimization state.

J.5 Transition Training Set

Before round t , the response model has access to

$$\mathcal{T}_t = \{(s_n, p_n, \mathbf{e}_n, \mathbf{m}_n)\}_{n=1}^{N_t}.$$

Each transition records the state from which an edit was evaluated, the candidate edit, its measured paired response, and the validity mask for that response.

An evaluated candidate remains in $\mathcal{T}_t$ regardless of whether it was subsequently committed. This allows the model to learn from unsuccessful or unsafe edits without allowing those edits to modify the deployed scoring specification.

J.6 Masked Response Loss

Let

$$\mu_{\theta,k}(s, p)$$

and

$$e_k$$

denote the k th predicted and observed response coordinates.

The response model minimizes

$$\mathcal{L}(\theta) = \sum_{(s,p,\mathbf{e},\mathbf{m}) \in \mathcal{T}_t} \sum_{k=1}^{d_e} m_k \ell(\mu_{\theta,k}(s, p), e_k).$$

The mask removes coordinates whose empirical repair or regression denominator is undefined. The component loss ℓ operates only on valid coordinates.

Because all coordinates of $\mathbf{e}_t(p)$ lie in $[0, 1]$, the sigmoid output constrains model predictions to the same range.

J.7 Warm-Start Transitions

The model is initialized from paired edit-effect measurements on $\mathcal{D}_{\text{dyn}, \text{train}}$, collected before model-guided selection. The number of answer–boundary observations is

$$N_{\text{boundary}} = \sum_{(s,p) \in \mathcal{T}_0} n_{s,p}(K_{s,p} - 1),$$

where $n_{s,p}$ counts complete answer–edit records for a measured state–edit intervention. Under a balanced design, this becomes $N_{\text{state}} N_{\text{edits/state}} N_{\text{answers/edit}} (K - 1)$. Our six-state pilot collected 36,000 boundary records, corresponding to 7,200 complete answer–edit records for the five-boundary scale. Boundary records are not treated as independent rubric trajectories. The additional observations from selected edits, including rejected edits, use the same training schema; $\mathcal{D}_{\text{cert}}$ and $\mathcal{D}_{\text{test}}$ are never used to fit the model.

State-held-out evaluation. Using rubric_state_id-grouped folds, the full state-conditioned model reaches $R^2 = 0.283$ on unseen states, compared with 0.241 for the edit-only model (a difference of 0.042). The separate edit_id and edit_text groupings check edit-instance and unseen-clause generalization; folds do not replace the four dataset partitions defined above.

J.8 Predictive Uncertainty

We train $E = 5$ independently seeded edit-effect models, following deep-ensemble uncertainty estimation (Lakshminarayanan et al., 2017). For candidate p_t^j , let $\mu_e(s_t, p_t^j)$ denote the prediction of ensemble member e . The ensemble mean and per-coordinate sample standard deviation are

$$\bar{\mu} = \frac{1}{E} \sum_{e=1}^E \mu_e,$$

$$\sigma = \sqrt{\frac{1}{E-1} \sum_{e=1}^E (\mu_e - \bar{\mu})^2}.$$

The selection rule uses $\bar{\mu}$ as its predicted mean; the standard deviation measures disagreement among models, not a calibrated confidence interval.

The corresponding target-repair quantities are

$$\mu_{t,j,b_t^*}^C, \quad \sigma_{t,j,b_t^*}^C,$$

while the regression quantities are

$$\mu_{t,j,b}^D, \quad \sigma_{t,j,b}^D.$$

These quantities are converted into

$$\tilde{C}_{t,j} = \mu_{t,j,b_t^*}^C - \beta \sigma_{t,j,b_t^*}^C,$$

and

$$\tilde{D}_{t,j,b} = \mu_{t,j,b}^D + \beta \sigma_{t,j,b}^D.$$

The predictive mean and standard deviation affect candidate ranking and filtering only. Even if the ensemble is miscalibrated, it cannot bypass the separate certification gate: commits rely on fresh $\mathcal{D}_{\text{cert}}$ observations and one-sided finite-sample bounds, subject to their sampling assumptions. Miscalibration can impair which candidates are explored; it does not itself provide authority to commit an edit.

J.9 Online Model Updating

After selecting p_t^* , the candidate is evaluated on $\mathcal{D}_{\text{dyn,train}}$ to obtain

$$\mathbf{e}_t^{\text{obs}}, \quad \mathbf{m}_t^{\text{obs}}.$$

The response history becomes

$$\mathcal{H}_{t+1} = \mathcal{H}_t \cup \left\{ (p_t^*, \mathbf{e}_t^{\text{obs}}, \mathbf{m}_t^{\text{obs}}) \right\},$$

and the supervised transition set becomes

$$\mathcal{T}_{t+1} = \mathcal{T}_t \cup \left\{ (s_t, p_t^*, \mathbf{e}_t^{\text{obs}}, \mathbf{m}_t^{\text{obs}}) \right\}.$$

The response model is updated from $\mathcal{T}_{t+1}$ before candidate selection at round $t + 1$.

If no candidate satisfies the predicted constraints at round t , no new paired transition is produced and

$$\mathcal{H}_{t+1} = \mathcal{H}_t, \quad \mathcal{T}_{t+1} = \mathcal{T}_t.$$

K Candidate Selection Details

This section expands the model-guided selection rule in Section 3.4.

K.1 Conservative Predicted Response

For candidate p_t^j targeting boundary b_t^* , define

$$\tilde{C}_{t,j} = \mu_{t,j,b_t^*}^C - \beta \sigma_{t,j,b_t^*}^C,$$

and

$$\tilde{D}_{t,j,b} = \mu_{t,j,b}^D + \beta \sigma_{t,j,b}^D.$$

The candidate chosen by the model is

$$p_t^* = \arg \max_{p_t^j \in \mathcal{P}_t} \tilde{C}_{t,j}$$

subject to

$$\tilde{D}_{t,j,b} \leq \epsilon_b, \quad \forall b \in \mathcal{B}.$$

The subtraction on repair penalizes uncertain predicted benefit. The addition on regression penalizes uncertain predicted harm.

These quantities are used only to determine which candidate should be measured with the frozen grader. They do not determine whether that candidate is committed.

K.2 No-Feasible-Candidate Case

If

$$\left\{ p_t^j \in \mathcal{P}_t : \tilde{D}_{t,j,b} \leq \epsilon_b \forall b \in \mathcal{B} \right\} = \emptyset,$$

the round produces no selected candidate.

In this case,

$$R_{t+1} = R_t,$$

and no new transition is added to the response history.

The optimizer therefore does not relax its predicted regression constraints merely to force an update at every round.

L Finite-Sample Certification

This section describes the statistical procedure used after a candidate has been selected and measured.

L.1 Certification Counts

At round t , candidate p_t^* is evaluated on the previously unused set

$$\mathcal{D}_{\text{cert}}^{(t)}.$$

For the target boundary b_t^* , let

$$n_t^C$$

be the number of certification examples that are incorrect before the candidate edit and therefore eligible for repair.

Let

$$x_t^C$$

be the number of those examples that become correct after the edit.

For each protected boundary b , let

$$n_{t,b}^D$$

be the number of examples that are correct before the edit, and let

$$x_{t,b}^D$$

be the number that become incorrect after the edit.

Conditional on the certification sampling distribution, the repair and regression indicators are treated as Bernoulli observations for the purpose of finite-sample confidence bounds.

L.2 Exact One-Sided Binomial Bounds

For an observed count x among n Bernoulli trials, we use exact one-sided binomial confidence bounds.

For target repair, the lower Clopper–Pearson bound is

$$L^C(x, n; \alpha_C) = \text{Beta}^{-1}(\alpha_C; x, n - x + 1)$$

for $x > 0$, with

$$L^C(0, n; \alpha_C) = 0.$$

At round t ,

$$L_{t,b_t}^C = L^C(x_t^C, n_t^C; \alpha_C).$$

For regression, the upper Clopper–Pearson bound is

$$U^D(x, n; \alpha_D) = \text{Beta}^{-1}(1 - \alpha_D; x + 1, n - x)$$

for $x < n$, with

$$U^D(n, n; \alpha_D) = 1.$$

Hence,

$$U_{t,b}^D = U^D(x_{t,b}^D, n_{t,b}^D; \alpha_{D,b}).$$

L.3 Certification Rule

A candidate can be certified only when

$$b_t^* \in \mathcal{B}_{\text{cert}}.$$

The repair requirement is

$$L_{t,b_t^*}^C \geq \gamma.$$

For every certified boundary,

$$U_{t,b}^D \leq \epsilon_b, \quad \forall b \in \mathcal{B}_{\text{cert}}.$$

A candidate is committed only when all required inequalities hold.

If the rule succeeds,

$$R_{t+1} = R_t \oplus p_t^*.$$

Otherwise,

$$R_{t+1} = R_t.$$

L.4 Confidence Allocation Across Multiple Bounds

At each round, the certificate contains $Q_{\text{cert}} = 1 + |\mathcal{B}_{\text{cert}}|$ one-sided statements: one for target correction and one for degradation at each certified boundary. Let δ be the family-wise error probability.

We allocate individual error probabilities

$$\alpha_1, \dots, \alpha_{Q_{\text{cert}}}$$

such that

$$\sum_{q=1}^{Q_{\text{cert}}} \alpha_q \leq \delta.$$

We use a Bonferroni allocation to control simultaneous coverage, as in classical family-wise error control (Holm, 1979). A simple Bonferroni allocation sets

$$\alpha_q = \frac{\delta}{Q_{\text{cert}}}.$$

Under this construction, the probability that at least one reported bound fails to cover its corresponding population quantity is at most δ under the assumed sampling model.

A per-edit certificate allocates confidence only over the statements included in that edit’s certificate. If a joint claim is made across multiple optimization rounds, the confidence allocation must additionally account for the statements made across those rounds.

L.5 Minimum Sample Support

A required quantity is classified as *unsupported* when its sample size is insufficient to establish the prescribed bound, even under the most favorable possible observation.

For regression, consider the favorable case in which zero harmful transitions are observed:

$$x = 0.$$

The exact upper bound reduces to

$$U^D(0, n; \alpha) = 1 - \alpha^{1/n}.$$

To certify

$$D \leq \epsilon$$

with zero observed regressions, n must satisfy

$$1 - \alpha^{1/n} \leq \epsilon.$$

Equivalently,

$$n \geq \frac{\log \alpha}{\log(1 - \epsilon)}.$$

We therefore define

$$n_{\min}^D = \left\lceil \frac{\log \alpha}{\log(1 - \epsilon)} \right\rceil.$$

A regression coordinate with

$$n_{t,b}^D < n_{\min}^D$$

cannot receive the requested certificate and is marked unsupported.

A corresponding minimum-support condition is applied to target repair: the available number of target errors must be large enough that even a perfect observed repair rate can produce

$$L^C \geq \gamma.$$

Illustrative sample-size calculation. For example, consider a regression tolerance

$$\epsilon = 0.08$$

with total error probability

$$\delta = 0.05$$

distributed equally across four simultaneous regression statements. Then

$$\alpha = \frac{0.05}{4} = 0.0125,$$

and zero observed regressions require

$$n_{\min}^D = \left\lceil \frac{\log(0.0125)}{\log(0.92)} \right\rceil = 53.$$

If the same error budget is instead allocated jointly over four boundaries and five rounds using a simple Bonferroni correction,

$$\alpha = \frac{0.05}{4 \times 5} = 0.0025,$$

which increases the corresponding zero-regression requirement to

$$n_{\min}^D = 72.$$

These calculations are sample-size illustrations for the stated numbers of regression bounds. Under the actual per-edit rule, the confidence allocation also includes the target-correction bound, giving $Q_{\text{cert}} = 1 + |\mathcal{B}_{\text{cert}}|$ statements when all certified boundaries contribute one degradation constraint. A joint multi-round claim would require the corresponding allocation across rounds. These calculations illustrate why an observed regression rate alone does not establish the prespecified finite-sample bounds.

L.6 Unsupported Outcomes

Certification produces three conceptually distinct outcomes.

Certified. All required repair and regression bounds are supported and satisfy the specified thresholds.

Rejected. The required quantities have sufficient support, but at least one confidence bound violates its threshold.

Unsupported. The target boundary lies outside the prespecified certification scope, or at least one required quantity lacks sufficient sample support for the prescribed bound.

Unsupported outcomes are never counted as certified. They are reported separately from statistical rejection because the two outcomes represent different reasons for withholding an update.

L.7 Per-Edit and Trajectory-Level Guarantees

A per-edit certificate applies only to the selected candidate and the fresh certification sample used at that round. It does not automatically provide a simultaneous guarantee for the entire optimization trajectory.

A joint multi-round guarantee requires confidence allocation over all statements included in that claim. Our round-wise certification splits permit such a construction because the certification observations for different rounds are separated in advance.

When an experiment does not use the required fresh certification data or joint confidence allocation, long-horizon results are described as empirical trajectory analyses rather than finite-sample certified guarantees.

L.8 Scope and Interpretation

The certification procedure addresses one narrow question: whether the observed repair and regression rates provide sufficient finite-sample evidence for the prescribed thresholds.

The certificate is conditional on the distribution from which the certification examples are sampled and on the reference labels used to define correctness.

It does not by itself establish educational validity, fairness, distribution-shift robustness, or the absence of all possible future grading errors.

The response model and certification procedure therefore play different roles. The former predicts which candidate is promising enough to evaluate; the latter determines whether the measured candidate has sufficient held-out evidence to be committed.

M Complete Optimization-State Update

For completeness, this section states the state update performed after each optimization round.

If no candidate satisfies the predicted selection constraints,

$$R_{t+1} = R_t,$$

$$\mathcal{H}_{t+1} = \mathcal{H}_t,$$

and

$$\mathcal{T}_{t+1} = \mathcal{T}_t.$$

If a candidate p_t^* is selected, it is first evaluated on $\mathcal{D}_{\text{dyn,train}}$. Its observed transition is added to the history and supervised transition data regardless of the later certification outcome:

$$\mathcal{H}_{t+1} = \mathcal{H}_t \cup \{(p_t^*, \mathbf{e}_t^{\text{obs}}, \mathbf{m}_t^{\text{obs}})\},$$

$$\mathcal{T}_{t+1} = \mathcal{T}_t \cup \{(s_t, p_t^*, \mathbf{e}_t^{\text{obs}}, \mathbf{m}_t^{\text{obs}})\}.$$

The certification outcome affects only the deployed scoring specification:

$$R_{t+1} = \begin{cases} R_t \oplus p_t^*, & \text{if } p_t^* \text{ is certified,} \\ R_t, & \text{otherwise.} \end{cases}$$

Thus, an edit may contribute information to the learned response model without being incorporated into the grader itself. This separation allows rejected transitions to improve future prediction while preventing uncertified edits from modifying the active rubric.

N Evaluation Metric Details

N.1 Edit-Response Prediction

Let $\mathcal{V}$ denote the held-out collection of measured state–edit transitions used for response-model evaluation. For transition (t, p) and boundary b , let $m_{t,b}^C$ and $m_{t,b}^D$ denote the validity indicators defined in Appendix I.3.

Let $q = (t, p, b)$ index the held-out state–edit–boundary set $\mathcal{I} = \{(t, p, b) : (t, p) \in \mathcal{V}, b \in \mathcal{B}\}$. Write $C_q = C_{t,b}(p)$, $D_q = D_{t,b}(p)$, and use the

same shorthand for predictions and validity masks. Then

$$\text{MAE}_C [\text{pp}] = 100 \frac{\sum_{q \in \mathcal{I}} m_q^C |\hat{C}_q - C_q|}{\sum_{q \in \mathcal{I}} m_q^C},$$

$$\text{MAE}_D [\text{pp}] = 100 \frac{\sum_{q \in \mathcal{I}} m_q^D |\hat{D}_q - D_q|}{\sum_{q \in \mathcal{I}} m_q^D}.$$

The factor of 100 reports absolute errors in percentage points, as in Section 4.2; the underlying correction and degradation rates themselves range from 0 to 1. Undefined response coordinates are removed through the corresponding validity indicators rather than assigned an artificial value.

N.2 Candidate-Selection Metrics

Let $\mathcal{S}$ denote the optimization rounds in which a candidate edit is selected and evaluated.

The realized target repair is

$$\text{Repair}_{\text{sel}} = \frac{1}{|\mathcal{S}|} \sum_{t \in \mathcal{S}} C_{t,b_t^*}(p_t^*).$$

Mean realized regression is

$$\text{Regression}_{\text{sel}} = \frac{\sum_{t \in \mathcal{S}} \sum_{b \in \mathcal{B}} m_{t,b}^D D_{t,b}(p_t^*)}{\sum_{t \in \mathcal{S}} \sum_{b \in \mathcal{B}} m_{t,b}^D}.$$

An edit is empirically feasible when every observed degradation coordinate satisfies its tolerance. Let $\mathcal{B}_t^D = \{b \in \mathcal{B} : m_{t,b}^D = 1\}$, and define

$$F_t(p) = \mathbb{I}[\mathcal{B}_t^D \neq \emptyset] \prod_{b \in \mathcal{B}_t^D} \mathbb{I}[D_{t,b}(p) \leq \epsilon_b].$$

The empirical feasibility rate is

$$\text{EFR} = \frac{1}{|\mathcal{S}|} \sum_{t \in \mathcal{S}} F_t(p_t^*).$$

EFR is an empirical selection metric. It should not be interpreted as a finite-sample certification guarantee; formal certification is defined separately in Appendix L.

O Prediction Variants and Ablation Controls

O.1 Four Predictor Variants

Figure 2(a) compares four ways of using the same candidate edit p . We write $a_p = \phi_{\text{hybrid}}(p)$ for the hybrid text–D2L edit encoding, $h_R = \psi_R(R_t)$ for the current rubric alone, and $h_s = \psi(R_t, \mathcal{H}_t, \mathbf{v}_t)$

Algorithm 1: RUBRICWORLD: State-Conditioned Rubric Optimization

Input: R_0 ; $\mathcal{D}_{\text{dyn},\text{train}}$; $\mathcal{D}_{\text{dyn},\text{val}}$; $\mathcal{D}_{\text{cert}}$; rounds T

Output: final specification R_T

```

1 Initialize the response model from warm-start
  transitions;
2 for  $t = 0, \dots, T - 1$  do
3   Generate candidate edits  $\mathcal{P}_t$ ;
4   Form the current state  $s_t$ ;
5   Predict edit responses and uncertainty;
6   Select  $p_t^*$  using Eq. (6);
7   if a candidate is selected then
8     Measure its paired response on  $\mathcal{D}_{\text{dyn},\text{train}}$ ;
9     Update the transition history and response
      model;
10    Compute certification bounds on held-out
      data;
11    if Eq. (7) holds then
12       $R_{t+1} \leftarrow R_t \oplus p_t^*$ ;
13    else
14       $R_{t+1} \leftarrow R_t$ ;
15  else
16     $R_{t+1} \leftarrow R_t$ ;
17 return  $R_T$ ;

```

for the full state (rubric text, measured edit history, and training-answer behavior). Every model applies the same element-wise sigmoid to its response logits and predicts the correction/degradation coordinates with the same masked training targets:

$$\begin{aligned}
 \eta_{\text{edit}} &= W_a a_p + c, \\
 \eta_{\text{rubric}} &= W_a a_p + W_R h_R + c, \\
 \eta_{\text{history}} &= W_a a_p + W_s h_s + c, \\
 \eta_{\text{full}} &= W_a a_p + U[(A a_p) \odot (B h_s)] + c.
 \end{aligned}$$

The corresponding labels in Figure 2(a) are *Edit only*, *Edit + current rubric*, *Edit + rubric + history*, and *RubricWorld (Ours)*. The third model also reads $\mathbf{v}_t$ as part of its full-state summary; “history” in its short plot label therefore includes the observed grading profile. The first three variants use additive input combinations; the full variant learns a factorized edit–state interaction. This comparison varies architecture while holding data splits, response masks, edit representations, and evaluation metrics fixed.

O.2 End-to-End Controls and Cost Accounting

In Table 2, *Full RubricWorld* uses the state-conditioned predictor for candidate selection. The *w/o State* variant retains learned prediction and the same hybrid edit representation but does not condition on the rubric state. The *w/o Learned Model*

variant replaces predicted effects with directly measured paired candidate effects when selecting edits; all variants retain the same candidate generator and independent certification rule.

Five-round quality is measured after five refinements. The 20-round column reports only the cumulative cost of continuing the same per-round procedure for fifteen more rounds: grader, candidate budget, data roles, and certification rule are unchanged. Token counts include warm-start collection when applicable, proposal generation, paired grading, and commit checks; final test evaluation is excluded. The quality–cost sweep in Figure 3 plots means over three random seeds, with each point including the warm-start tokens. Local predictor training and embedding computation are separate from LLM-token usage and are not represented as free wall-clock compute.